Stat 136 (Bayesian Statistics)

Lesson 3.1 Bayesian Regression Analysis

Learning Objectives

At the end of this lesson, students are expected to

1. articulate how to Bayesian regression differs from frequentist regression approach;
2. formulate a Bayesian regression model with specified priors and likelihood functions;
3. apply computational methods to estimate posterior distributions of the parameters of regression models; and
4. report and interpret Bayesian regression results.

Introduction

We have learned in the previous modules how to implement the Bayesian paradigm in different estimation and hypothesis testing scenarios.

In this lesson, we will first apply Bayesian statistics to simple linear regression models, then generalize the results to multiple linear regression models. We will see when using the reference prior, the posterior means, posterior standard deviations, and credible intervals of the coefficients coincide with the counterparts in the frequentist ordinary least square (OLS) linear regression models. However, using the Bayesian framework, we can now interpret credible intervals as the probabilities of the coefficients lying in such intervals.

Bayesian Simple Linear Regression

Frequentist Ordinary Least Square (OLS) Simple Linear Regression

Obtaining accurate measurements of body fat is expensive and not easy to be done. Instead, predictive models that predict the percentage of body fat which use readily available measurements such as abdominal circumference are easy to use and inexpensive. We will apply a simple linear regression to predict body fat using abdominal circumference as an example to illustrate the Bayesian approach of linear regression. The data set *bodyfat* can be found in the **BAS** package in R.

```r
#Loading the required packages and the data set
library(BAS)
library(tidyverse)
library(ggpubr)
data(bodyfat)
head(bodyfat)
```

```
##   Density Bodyfat Age Weight Height Neck Chest Abdomen   Hip Thigh Knee Ankle
## 1  1.0708    12.3  23 154.25  67.75 36.2  93.1    85.2  94.5  59.0 37.3  21.9
## 2  1.0853     6.1  22 173.25  72.25 38.5  93.6    83.0  98.7  58.7 37.3  23.4
## 3  1.0414    25.3  22 154.00  66.25 34.0  95.8    87.9  99.2  59.6 38.9  24.0
## 4  1.0751    10.4  26 184.75  72.25 37.4 101.8    86.4 101.2  60.1 37.3  22.8
## 5  1.0340    28.7  24 184.25  71.25 34.4  97.3   100.0 101.9  63.2 42.2  24.0
## 6  1.0502    20.9  24 210.25  74.75 39.0 104.5    94.4 107.8  66.0 42.0  25.6
##   Biceps Forearm Wrist
## 1   32.0    27.4  17.1
## 2   30.5    28.9  18.2
## 3   28.8    25.2  16.6
## 4   32.4    29.4  18.2
## 5   32.2    27.7  17.7
## 6   35.7    30.6  18.8
```

This data frame includes 252 observations of men's body fat and other measurements, such as waist circumference (Abdomen). We will construct a Bayesian model of simple linear regression, which uses *Abdomen* to predict the response variable *Bodyfat*.

This regression model can be formulated as

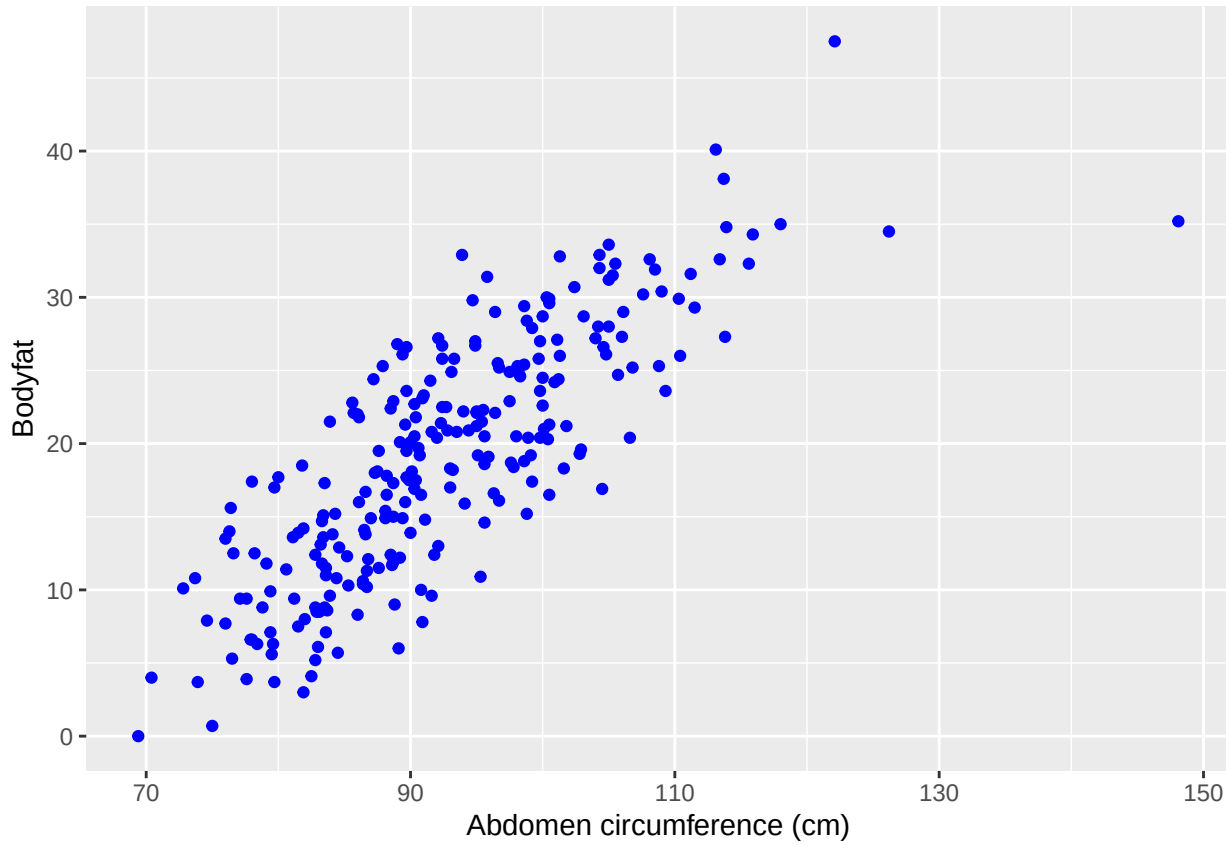
$$y_i = \alpha + \beta x_i + \epsilon_i,\ i = 1, 2, \cdots 252$$

Here, we assume error ϵ_i is independent and identically distributed as normal random variables with mean zero and constant variance σ^2:

$$\epsilon_i \sim N(0, \sigma^2).$$

The scatter plot of the percentage body fat and the abdominal circumference measurements for 252 men is shown below.

```r
ggplot(data = bodyfat, aes(x = Abdomen,
                           y = Bodyfat)) +
  geom_point(color = "blue") +
  xlab("Abdomen circumference (cm)")
```



A positive correlation between the variables *Abdomen* and *Bodyfat* is apparent in the scatter plot. To predict body fat, we fit an ordinary least squares (OLS) regression using the *lm()* function in R.

```
# Frequentist OLS linear regression
bodyfat.lm <-  lm(Bodyfat ~ Abdomen, data = bodyfat)
summary(bodyfat.lm)
```

```
##
## Call:
## lm(formula = Bodyfat ~ Abdomen, data = bodyfat)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.0160  -3.7557   0.0554   3.4215  12.9007
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -39.28018    2.66034  -14.77   <2e-16 ***
## Abdomen       0.63130    0.02855   22.11   <2e-16 ***
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.877 on 250 degrees of freedom
## Multiple R-squared:  0.6617, Adjusted R-squared:  0.6603
## F-statistic: 488.9 on 1 and 250 DF,  p-value: < 2.2e-16
```

From the summary, we see that this model has an estimated slope, $\hat{\beta} \approx 0.63$ and an estimated y-intercept, $\hat{\alpha} = -39.28$. This gives us the prediction formula

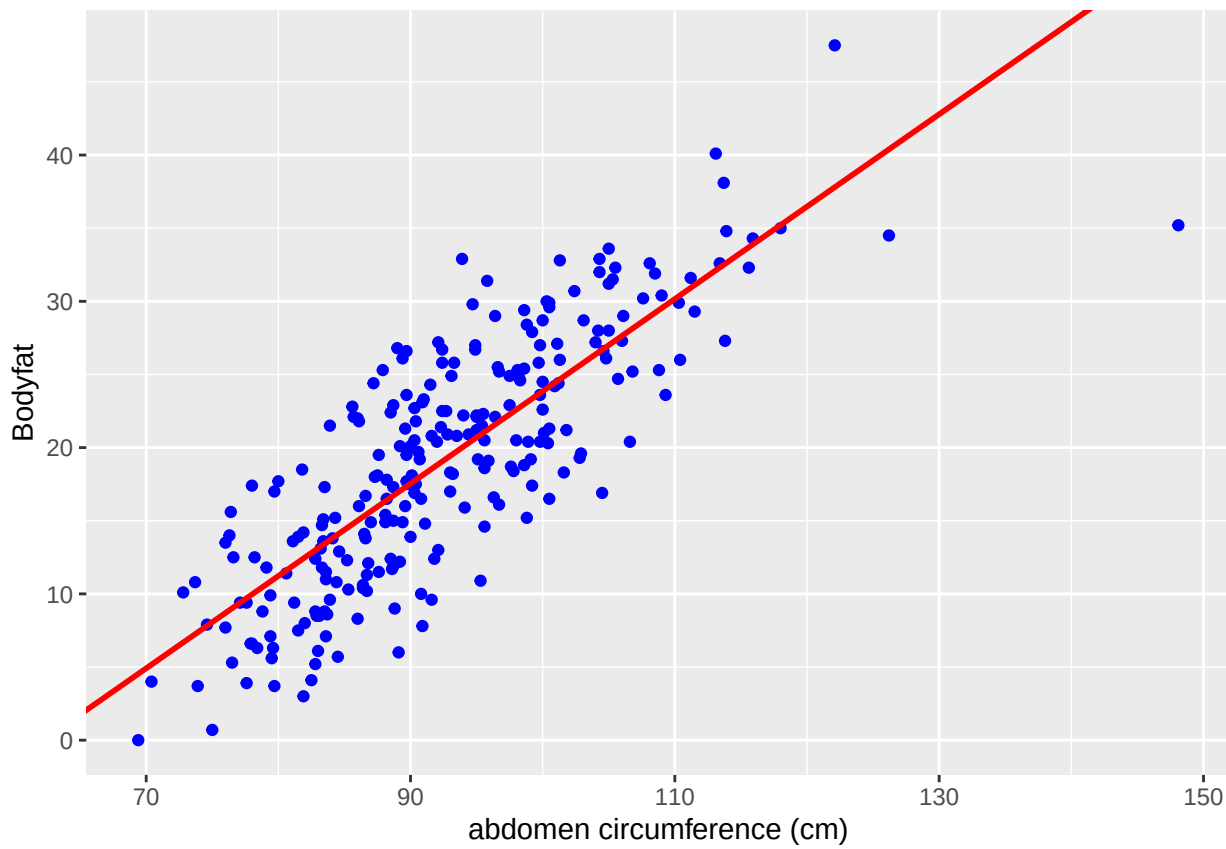
$$\widehat{Bodyfat} = -39.28 + 0.63 Abdomen$$

For every additional centimeter, we expect body fat to increase by 0.63%. The negative y-intercept of course does not make sense as a physical model, but neither does predicting a male with a waist of zero centimeters. Nevertheless, this linear regression may be an accurate approximation for prediction purposes for measurements that are in the observed range for this population.

We overlayed on the scatter plot the estimated regression line.

```r
# Extract coefficients
beta <-  coef(bodyfat.lm)

# Visualize regression line on the scatter plot
library(ggplot2)
ggplot(data = bodyfat, aes(x = Abdomen,
                           y = Bodyfat)) +
  geom_point(color = "blue") +
  geom_abline(intercept = beta[1],
              slope = beta[2],
              size = 1,
              color = "red") +
  xlab("abdomen circumference (cm)")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Each of the residuals, which provide an estimate of the fitting error (we call them the **residuals**), is equal to $\hat{\epsilon}_i = y_i - \hat{y}_i$, the difference between the observed value y_i and the fitted value $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$, where x_i is the abdominal circumference for the ith male.

The residuals $\hat{\epsilon}_i$ is used for diagnostics as well as estimating the constant variance in the assumption of the model σ^2 via the mean squared error (MSE):

$$\hat{\sigma^2} = \frac{1}{n-2}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n-2}\sum_{i=1}^{n}\hat{\epsilon}_i^2$$

Here the degrees of freedom $n-2$ are the number of observations adjusted for the number of parameters (which is 2) that we estimated in the regression. The MSE, $\hat{\sigma^2}$, may be calculated through squaring the residuals of the output of *bodyfat.lm*.

```
# Obtain residuals and n
resid = residuals(bodyfat.lm)
n = length(resid)

# Calculate MSE
MSE = 1/ (n - 2) * sum((resid ^ 2))
print(round(MSE,2))
```

```
## [1] 23.79
```

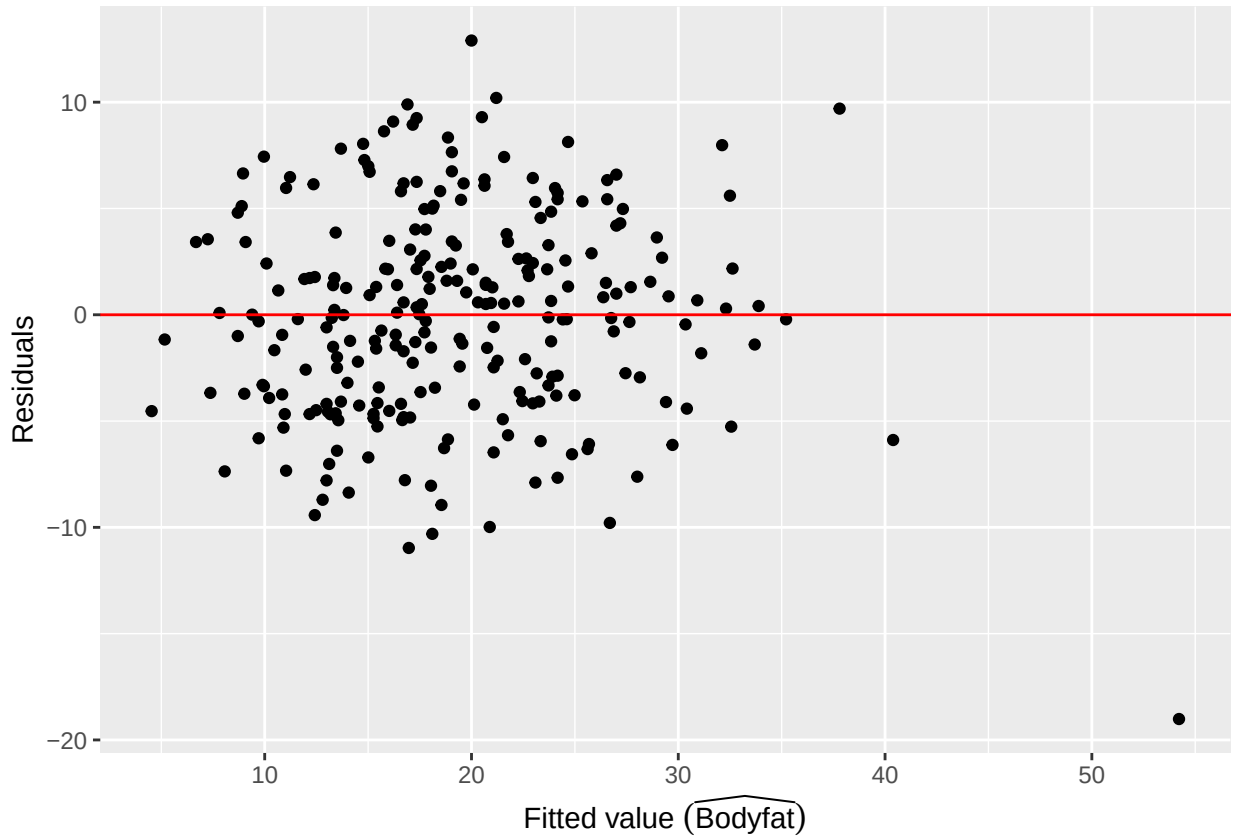
Alternatively, you can request for the ANOVA table for this regression output.

```r
options(digits = 8)
anova(bodyfat.lm)
```

```
## Analysis of Variance Table
##
## Response: Bodyfat
##            Df   Sum Sq  Mean Sq F value     Pr(>F)
## Abdomen     1 11631.53 11631.53 488.928 < 2.22e-16 ***
## Residuals 250  5947.46    23.79
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

If this model is correct, the residuals and fitted values should be uncorrelated, and the expected value of the residuals is zero. We apply the scatterplot of residuals versus fitted values, which provides an additional visual check of the model adequacy.

```r
# Combine residuals and fitted values into a data frame
result = data.frame(fitted_values = fitted.values(bodyfat.lm),
                    residuals = residuals(bodyfat.lm))

# Load library and plot residuals versus fitted values
library(ggplot2)
ggplot(data = result, aes(x = fitted_values,
                          y = residuals)) +
  geom_point() +
  geom_abline(intercept = 0, slope = 0, col = "red") +
  xlab(expression(paste("Fitted value ", (widehat(Bodyfat))))) +
  ylab("Residuals")
```



With the exception of one observation for the individual with the largest fitted value, the residual plot suggests that this linear regression is a reasonable approximation. The case number of the observation with the largest fitted value can be obtained using the *which()* function in R. Further examination of the data frame shows that this case also has the largest waist measurement Abdomen.

```
# Find the observation with the largest fitted value
which.max(as.vector(fitted.values(bodyfat.lm)))
```

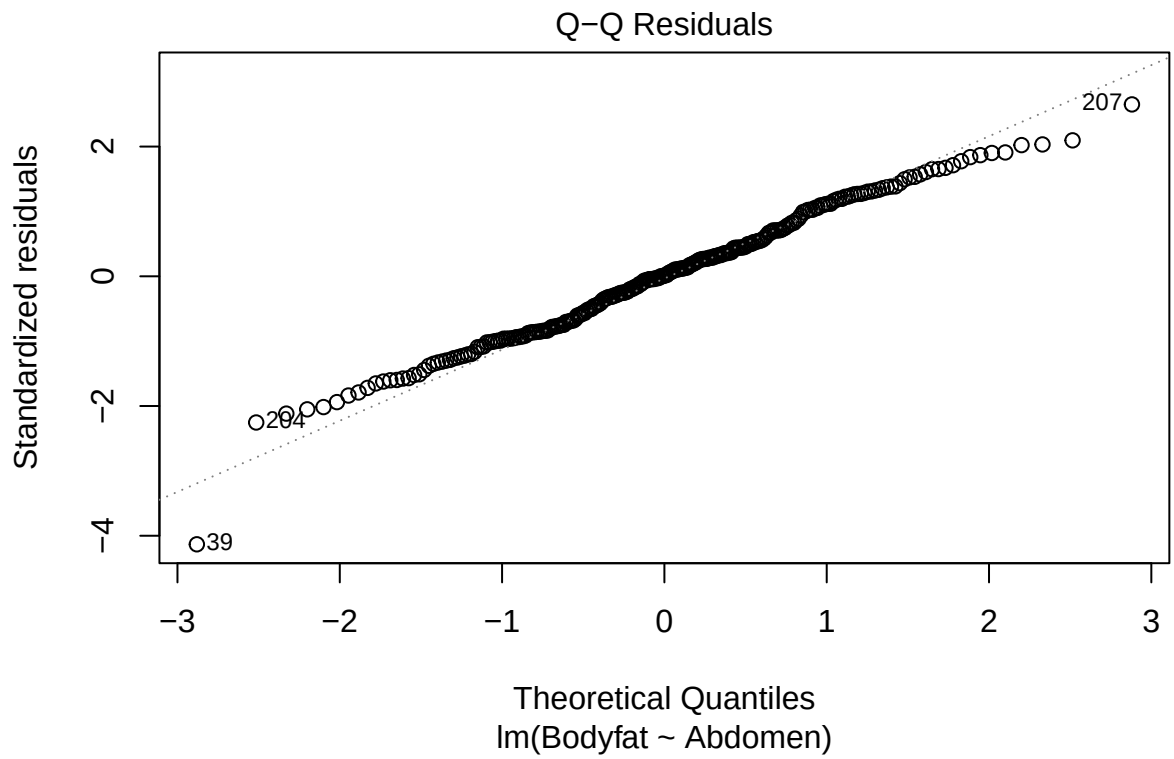
```
## [1] 39
```

```
# Shows this observation has the largest Abdomen
which.max(bodyfat$Abdomen)
```

```
## [1] 39
```

Furthermore, we can check the normal probability plot of the residuals for the assumption of normally distributed errors. We see that only Case 39, the one with the largest waist measurement, is exceptionally away from the normal quantile.

```r
plot(bodyfat.lm, which = 2) #which = 2 option requests the QQ plot
```



The confidence interval of α and β can be constructed using the standard errors $SE(\alpha)$ and $SE(\beta)$, respectively. To proceed, we introduce notations of some "sums of squares":

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2$$
$$S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2$$
$$S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$
$$SSE = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n}\hat{\epsilon}_i^2$$

The estimates of α and β, which are denoted as $\hat{\alpha}$ and $\hat{\beta}$, respectively, can be calculated using these "sums of squares".

$$\hat{\beta} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}$$

and

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = \bar{y} - \left(\frac{S_{xy}}{S_{xx}}\right)\bar{x}$$

The last "sum of square" is the sum of squares of errors (SSE). Its sample mean is exactly the mean squared error (MSE) we introduced previously

$$\hat{\sigma}^2 = \frac{SSE}{n-1} = MSE$$

The standard errors, $SE(\alpha)$ and $SE(\beta)$, are given as

$$SE(\alpha) = \sqrt{\frac{SSE}{n-2}\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)} = \hat{\sigma}\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$$

and

$$SE(\beta) = \sqrt{\frac{SSE}{n-2}\left(\frac{1}{S_{xx}}\right)} = \frac{\hat{\sigma}}{\sqrt{S_{xx}}}$$

We may construct the confidence intervals of α and β using the following t statistics:

$$t_\alpha = \frac{\alpha - \hat{\alpha}}{SE(\alpha)}$$

and

$$t_\beta = \frac{\beta - \hat{\beta}}{SE(\beta)}$$

Both statistics have $n-2$ degrees of freedom. Therefore, a $(1-\alpha)$ confidence interval for α and β is constructed as follows:

$$\hat{\alpha} \pm t_{\alpha/2,n-2}SE(\hat{\alpha})$$

and

$$\hat{\beta} \pm t_{\alpha/2,n-2}SE(\hat{\beta})$$

For the given example, a 95% confidence intervals for α and β is constructed as follows:

$$\hat{\alpha} \pm t_{0.025,n-2} SE(\hat{\alpha}) \implies (-39.28018 \pm 1.97(2.66034)) \implies (-44.52, -34.04)$$

and

$$\hat{\beta} \pm t_{0.025,n-2} SE(\hat{\beta}) \implies (0.63130 \pm 1.97(0.02855)) \implies (0.58, 0.69)$$

We can confirm these 95% confidence intervals using the following code chunk.

```
confint(bodyfat.lm)
```

```
##                    2.5 %       97.5 %
## (Intercept) -44.51971403 -34.04065528
## Abdomen       0.57507388   0.68753488
```

Bayesian Simple Linear Regression Using the Reference Prior

Let us now turn to the Bayesian version and show that under the reference prior, we will obtain the posterior distributions of α and β analogous with the frequentist OLS results.

The Bayesian model starts with the same model as the classical frequentist approach:

$$y_i = \alpha + \beta x_i + \epsilon_i, \;\; i = 1, 2, \cdots n$$

with the assumption that the errors, ϵ_i, are independent and identically distributed as normal random variables with mean zero and constant variance σ^2. This assumption is exactly the same as in the classical inference case for testing and constructing confidence intervals for α and β.

Our goal is to update the distributions of the unknown parameters α, β, and σ^2, based on the data $(x_1, y_1), (x_2, y_2), \cdots (x_n, y_n)$, where n is the number of observations.

Under the assumption that the errors ϵ_i are normally distributed with constant variance σ^2, we have for the random variable of each response Y_i, conditioning on the observed data x_i and the parameters α, β, and σ^2, is normally distributed:

$$Y_i | x_i, \alpha, \beta, \sigma^2 \sim N(\alpha + \beta x_i, \sigma^2), \;\; i = 1, 2, \cdots, n.$$

That is, the likelihood of each Y_i given x_i, α, β, and σ^2 is given by

$$f_Y(y_i | x_i, \alpha, \beta, \sigma^2) = \frac{1}{\sigma\sqrt{2\pi}} exp\left[-\frac{1}{2\sigma^2} \left(y_i - (\alpha + \beta x_i)\right)^2 \right]$$

The likelihood of $Y_1, Y_2, \cdots, Y_n$ is the product of each likelihood $f_Y(y_i | x_i, \alpha, \beta, \sigma^2)$, since we assume each response Y_i is independent from each other. Since this likelihood depends on the values of α, β, and σ^2, it is sometimes denoted as a function of α, β, and σ^2: $\mathbf{L}(\mathbf{Y} | \alpha, \beta, \sigma^2)$.

We first consider the case under the reference prior, which is our standard noninformative prior. Using the reference prior, we will obtain familiar distributions as the posterior distributions of α, β, and σ^2, which gives the analogue to the frequentist results. Here we assume the joint prior distribution of α and β as uniform, that is,

$$\pi(\alpha, \beta) \propto 1$$

and the prior of σ^2 to be proportional to the inverse of σ^2:

$$\pi(\sigma^2) \propto \frac{1}{\sigma^2}.$$

Combining these two we get the joint prior distribution of α, β, and σ^2:

$$\pi(\alpha, \beta, \sigma^2) \propto \frac{1}{\sigma^2}$$

Then we apply the Bayes' rule to derive the joint posterior distribution after observing data $y_1, y_2, \cdots, y_n$. Bayes' rule states that the joint posterior distribution of α, β, and σ^2 is proportional to the product of the likelihood and the joint prior distribution:

$$\begin{aligned}
\pi(\alpha, \beta, \sigma^2 | y_1, y_2, \cdots, y_n) &\propto \left[\prod_{i=1}^{n} f_Y(y_i | x_i, \alpha, \beta, \sigma^2) \right] \times \pi(\alpha, \beta, \sigma^2) \\
&= \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} exp\left[-\frac{1}{2\sigma^2} \left(y_i - (\alpha + \beta x_i)\right)^2 \right] \times \frac{1}{\sigma^2} \\
&\propto \frac{1}{(\sigma^2)^{(n+2)/2}} exp\left[-\frac{1}{2\sigma^2} \sum (y_i - \alpha_i - \beta x_i)^2 \right].
\end{aligned}$$

To obtain the marginal posterior distribution of β, we need to integrate α and σ^2 out from the joint posterior distribution. That is,

$$\pi(\beta | y_1, y_2, \cdots, y_n) = \int_0^\infty \left(\int_{-\infty}^{\infty} \pi(\alpha, \beta, \sigma^2 | y_1, y_2, \cdots, y_n) \, d\alpha \right) \, d\sigma^2$$

It can be shown that the marginal posterior distribution of β is the Student's t distribution.

$$\beta | y_1, y_2, \cdots, y_n \sim t\Big[n - 2, \hat{\beta}, SE(\beta)^2\Big]$$

Similarly, we can integrate out β and σ^2 from the joint posterior distribution to get the marginal posterior distribution of α. It turns out that

$$\alpha | y_1, y_2, \cdots, y_n \sim t\Big[n - 2, \hat{\alpha}, SE(\alpha)^2\Big]$$

Finally, we can show that the marginal posterior distribution of σ^2 is the inverse Gamma distribution, or equivalently, the precision $\phi = \frac{1}{\sigma^2}$ follows the Gamma distribution. That is,

$$\phi = \frac{1}{\sigma^2} \sim Gamma\left(\frac{n-2}{2}, \frac{SSE}{2}\right)$$

Moreover, using a Normal-Gamma conjugacy under the reference prior, the joint posterior distribution of β and σ^2, and the joint posterior distribution of α and σ^2 are both Normal-Gamma. In particular, the posterior distribution of β conditioning on σ^2 is

$$\beta|\sigma^2, data \sim N\left(\hat{\beta}, \frac{\sigma^2}{S_{xx}}\right)$$

and the posterior distribution of α conditioning on σ^2 is

$$\alpha|\sigma^2, data \sim N\left(\hat{\alpha}, \frac{\sigma^2}{n} + \frac{\sigma^2\bar{x}^2}{S_{xx}}\right)$$

Credible Intervals for α and β

The Bayesian posterior distribution results of α and β show that under the reference prior, the posterior credible intervals are in fact numerically equivalent to the confidence intervals from the classical frequentist OLS analysis. This provides a baseline analysis for other Bayesian analyses with other informative prior distributions or perhaps other "objective" prior distributions, such as the Cauchy distribution. (Cauchy distribution is the Student's t prior with 1 degree of freedom.)

```r
library(rstanarm)
bayes.lm <- stan_glm(Bodyfat ~ Abdomen,
                     family = gaussian(),
                     data = bodyfat,
                     set.seed(1234),
                     refresh = 0

)

coef(bayes.lm)
```

```
##  (Intercept)      Abdomen
## -39.34601130   0.63221241
```

```r
posterior_interval(bayes.lm)
```

```
##                        5%          95%
## (Intercept) -43.69806551 -34.93053997
## Abdomen       0.58460633   0.67915222
## sigma         4.54186133   5.27462728
```

```r
library(INLA)
formula <- Bodyfat ~ Abdomen
bayes.lm1 <- inla(formula, family="gaussian", data=bodyfat)
summary(bayes.lm1)
```

```
## Time used:
##     Pre = 1.33, Running = 0.279, Post = 0.0874, Total = 1.7
## Fixed effects:
##                mean    sd 0.025quant 0.5quant 0.975quant    mode kld
## (Intercept) -39.280 2.659    -44.499  -39.280    -34.061 -39.280   0
## Abdomen       0.631 0.029      0.575    0.631      0.687   0.631   0
##
## Model hyperparameters:
##                                          mean    sd 0.025quant 0.5quant
## Precision for the Gaussian observations 0.042 0.004      0.035    0.042
##                                         0.975quant  mode
## Precision for the Gaussian observations       0.05 0.042
##
## Marginal log-Likelihood:  -777.74
##  is computed
## Posterior summaries for the linear predictor and the fitted values are computed
## (Posterior marginals needs also 'control.compute=list(return.marginals.predictor=TRUE
```

```r
cbind(bayes.lm1$summary.fixed[,1:2],summary(bodyfat.lm)$coef[,1:2])
```

```
##                     mean          sd     Estimate  Std. Error
## (Intercept) -39.28013707 2.659198409 -39.28018466 2.660336964
## Abdomen       0.63130387 0.028538451   0.63130438 0.028550671
```

Credible Intervals for the Mean μ_Y and the Prediction y_{n+1}

From our assumption of the model

$$y_i = \alpha + \beta x_i + \epsilon_i$$

the mean of the response variable μ_Y at the point x_i is

$$\mu_{Y|x_i} = E[Y|x_i] = \alpha + \beta x_i.$$

Under the reference prior, μ_Y has a posterior distributuion

$$\alpha + \beta x_i | data \sim t(n-2, \hat{\alpha} + \hat{\beta} x_i, S^2_{Y|x_i})$$

where

$$S^2_{Y|x_i} = \hat{\sigma}^2 \left(\frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}} \right)$$

On the other hand, any new prediction y_{n+1} at point x_{n+1} also follows the Student's t distribution.

$$y_{n+1} | data, x_{n+1} \sim t(n-2, \hat{\alpha} + \hat{\beta} x_{n+1}, S^2_{Y|x_{n+1}})$$

where

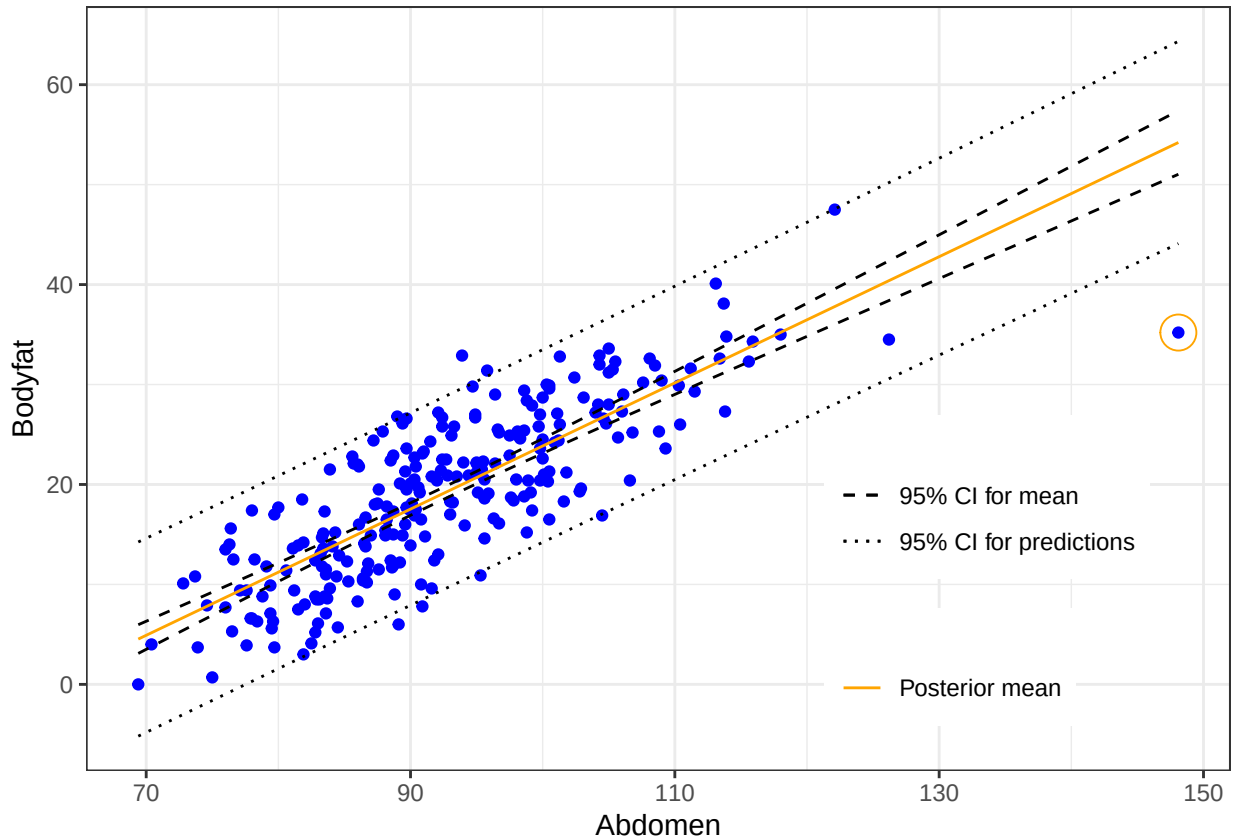
$$S^2_{Y|x_{n+1}} = \hat{\sigma}^2 \left(1 + \frac{1}{n} + \frac{(x_{n+1} - \bar{x})^2}{S_{xx}} \right)$$

This shows that the variance for predicting a new observation y_{n+1} has an extra $\hat{\sigma}^2$ which comes from the uncertainty of a new observation about the mean μ_Y estimated by the regression line.

We can extract these intervals using the *predict()* function.

```
#library(ggplot2)
# Construct current prediction
alpha = bodyfat.lm$coefficients[1]
beta = bodyfat.lm$coefficients[2]
new_x = seq(min(bodyfat$Abdomen), max(bodyfat$Abdomen),
            length.out = 100)
y_hat = alpha + beta * new_x

# Get lower and upper bounds for mean
ymean = data.frame(predict(bodyfat.lm,
                           newdata = data.frame(Abdomen = new_x),
                           interval = "confidence",
                           level = 0.95))

# Get lower and upper bounds for prediction
ypred = data.frame(predict(bodyfat.lm,
                          newdata = data.frame(Abdomen = new_x),
                          interval = "prediction",
                          level = 0.95))

output = data.frame(x = new_x, y_hat = y_hat, ymean_lwr = ymean$lwr, ymean_upr = ymean$u
                    ypred_lwr = ypred$lwr, ypred_upr = ypred$upr)
```

```r
# Extract potential outlier data point
outlier = data.frame(x = bodyfat$Abdomen[39],
                     y = bodyfat$Bodyfat[39])

# Scatter plot of original
plot1 = ggplot(data = bodyfat, aes(x = Abdomen,
                                   y = Bodyfat)) +
  geom_point(color = "blue")

# Add bounds of mean and prediction
plot2 = plot1 +
  geom_line(data = output, aes(x = new_x,
                               y = y_hat,
                               color = "first"),
                               lty = 1) +
  geom_line(data = output, aes(x = new_x,
                               y = ymean_lwr,
                               lty = "second")) +
  geom_line(data = output, aes(x = new_x,
                               y = ymean_upr,
                               lty = "second")) +
  geom_line(data = output, aes(x = new_x,
                               y = ypred_upr,
                               lty = "third")) +
  geom_line(data = output, aes(x = new_x,
                               y = ypred_lwr,
                               lty = "third")) +
  scale_colour_manual(values = c("orange"),
                      labels = "Posterior mean",
                      name = "") +
  scale_linetype_manual(values = c(2, 3),
                        labels = c("95% CI for mean", "95% CI for predictions"),
                        name = "") +
  theme_bw() +
  theme(legend.position = c(1.1, 0.1),
        legend.justification = c(1.6, 0.1))

# Identify potential outlier
plot2 + geom_point(data = outlier, aes(x = x, y = y),
                   color = "orange",
                   pch = 1,
                   cex = 6)
```



Note in the above plot, the legend "CI" can mean either confidence interval or credible interval. The difference comes down to the interpretation. For example, the prediction at the same abdominal circumference as in Case 39 is

```
pred.39 = predict(bodyfat.lm,
                  newdata = bodyfat[39, ],
                  interval = "prediction", level = 0.95)
out = cbind(bodyfat[39,]$Abdomen, pred.39)
colnames(out) = c("abdomen", "prediction", "lower", "upper")
out
```

```
##    abdomen prediction     lower     upper
## 39   148.1  54.215994 44.096705 64.335284
```

Based on the data, a Bayesian would expect that a man with waist circumference of 148.1 centimeters should have bodyfat of 54.216% with a 95% chance that it is between 44.097% and 64.335%.

While we expect the majority of the data will be within the prediction intervals (the short dashed grey lines), Case 39 seems to be well below the interval.

Informative Priors

Except from the noninformative reference prior, we may also consider using a more general semi-conjugate prior distribution of α, β, and σ^2 when there is information available about the parameters.